

Sampling Methods & Bias



Identifying sampling methods

- 1 A school assigns each student a number from 1 to 800, then uses a random number generator to select 40 numbers, surveying those students. Identify this sampling method.
- 2 A researcher divides a school into grade levels (9th, 10th, 11th, 12th), then takes a simple random sample of 20 students from each grade. Identify this sampling method and explain why it might be preferred over an SRS of the whole school.

Cluster sampling and systematic sampling

- 3 A researcher randomly selects 5 entire homeroom classes out of 40 in a school, and surveys every student in those 5 classes. Identify this sampling method, and contrast it with stratified sampling.
- 4 A quality-control inspector checks every 25th item coming off a production line. Identify this sampling method.

Sources of bias

- 5 A survey about a new school policy is voluntarily filled out online, and only 12% of students respond. Identify the type of bias most likely present, and explain why the results may not represent the whole student body.
- 6 A phone survey about internet usage habits is conducted using only landline phone numbers. Identify the type of bias this introduces.

Wording and response bias

- 7 A survey question reads: "Don't you agree that the wasteful new cafeteria policy should be reversed?" Explain why this question is problematic, and rewrite it in a neutral form.
- 8 Explain why respondents might not answer a sensitive question (such as about illegal activity) truthfully in a face-to-face interview, and name this type of bias.

Comparing sample size and bias

- 9 Explain why increasing the sample size does not fix a biased sampling method, using the voluntary response survey from earlier as an example.
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Designing a sampling plan

- 10 A city has 4 distinct neighborhoods with very different income levels, and the mayor wants to survey residents about a new tax proposal in a way that fairly represents each neighborhood. Recommend a sampling method, and describe how it would be carried out.
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Convenience sampling and its pitfalls

- 11 A researcher stands outside a gym and surveys the first 50 people who walk out about their exercise habits, intending to represent the whole city's exercise habits. Identify this sampling method and explain the flaw in generalizing its results to the whole city.
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Comparing sampling methods side by side

- 12 A national retailer wants to estimate average customer satisfaction across all 500 of its stores. Compare how a cluster sample (randomly selecting 20 whole stores and surveying all customers there) and a stratified sample (grouping stores by region, then sampling customers from every region) would each be carried out, and state one advantage of each approach.
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Identifying the population and sample

- 13 A university wants to know the average commute time for all 30,000 enrolled students, and surveys a random sample of 400 students. Identify the population and the sample in this scenario.
- 14 A national news outlet runs a text-in poll during a live broadcast, asking viewers to text "YES" or "NO" on a controversial policy question, and reports the results as representing "public opinion." Identify every source of bias you can find in this scenario, and explain why the reported percentages should not be trusted as an accurate reflection of the general public's views.
- 15 A hospital wants to estimate the average patient satisfaction score across its 12 different departments (e.g. cardiology, pediatrics, emergency), which are expected to have very different typical satisfaction levels. Describe, step by step, how you would design a stratified random sample to estimate overall patient satisfaction, and explain why a simple random sample of all patients might be less informative for comparing satisfaction across departments.

- 16** A researcher wants to estimate the average time it takes customers to be served at a chain of 200 coffee shops nationwide. Due to budget constraints, the researcher can only visit a small number of physical locations. Describe how a cluster sample could be carried out in this situation, and explain a key risk of cluster sampling that would not apply if the researcher instead used a stratified sample based on region.
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Multiple choice — easy 15

- 17** A factory assigns each worker a number from 1 to 1223, then uses a random number generator to select 70 numbers, surveying those workers. Identify this sampling method.
- a. Convenience sample
 - b. Simple random sample (SRS)
 - c. Stratified random sample
 - d. Cluster sample
- 18** A researcher divides a factory into groups (morning shift, afternoon shift, and night shift), then takes a simple random sample of 53 workers from each group. Identify this sampling method.
- a. Cluster sample
 - b. Convenience sample
 - c. Stratified random sample
 - d. Systematic sample
- 19** A researcher randomly selects 3 entire delivery routes out of 33 at a shipping company, and surveys every member within those selected delivery routes. Identify this sampling method.
- a. Stratified random sample
 - b. Simple random sample (SRS)
 - c. Systematic sample
 - d. Cluster sample

- 20** A quality-control inspector at a factory checks every 49th item coming off a production line, starting from a randomly chosen point among the first 49 items. Identify this sampling method.
- a.** Simple random sample (SRS)
 - b.** Convenience sample
 - c.** Systematic sample
 - d.** Stratified random sample
- 21** A researcher stands outside a farmers market and surveys the first 44 people encountered, intending to represent the general population. Identify this sampling method.
- a.** Systematic sample
 - b.** Cluster sample
 - c.** Simple random sample (SRS)
 - d.** Convenience sample
- 22** A company assigns each employee a number from 1 to 1017, then uses a random number generator to select 61 numbers, surveying those employees. Identify this sampling method.
- a.** Simple random sample (SRS)
 - b.** Convenience sample
 - c.** Cluster sample
 - d.** Systematic sample
- 23** A researcher divides a hospital into groups (department A, department B, and department C), then takes a simple random sample of 72 patients from each group. Identify this sampling method.
- a.** Cluster sample
 - b.** Stratified random sample
 - c.** Convenience sample
 - d.** Simple random sample (SRS)

- 24** A researcher randomly selects 6 entire wards out of 18 at a hospital, and surveys every member within those selected wards. Identify this sampling method.
- a.** Stratified random sample
 - b.** Convenience sample
 - c.** Systematic sample
 - d.** Cluster sample
- 25** A quality-control inspector at a factory checks every 44th item coming off a production line, starting from a randomly chosen point among the first 44 items. Identify this sampling method.
- a.** Simple random sample (SRS)
 - b.** Stratified random sample
 - c.** Convenience sample
 - d.** Systematic sample
- 26** A researcher stands outside a train station and surveys the first 49 people encountered, intending to represent the general population. Identify this sampling method.
- a.** Cluster sample
 - b.** Simple random sample (SRS)
 - c.** Convenience sample
 - d.** Systematic sample
- 27** A hospital assigns each patient a number from 1 to 1075, then uses a random number generator to select 80 numbers, surveying those patients. Identify this sampling method.
- a.** Systematic sample
 - b.** Simple random sample (SRS)
 - c.** Stratified random sample
 - d.** Cluster sample

- 28** A researcher divides a factory into groups (freshmen, sophomores, juniors, and seniors), then takes a simple random sample of 67 workers from each group. Identify this sampling method.
- a.** Cluster sample
 - b.** Systematic sample
 - c.** Stratified random sample
 - d.** Simple random sample (SRS)
- 29** A researcher randomly selects 9 entire homeroom classes out of 15 at a school, and surveys every member within those selected homeroom classes. Identify this sampling method.
- a.** Simple random sample (SRS)
 - b.** Systematic sample
 - c.** Cluster sample
 - d.** Convenience sample
- 30** A quality-control inspector at a clinic checks every 32nd item coming off a production line, starting from a randomly chosen point among the first 32 items. Identify this sampling method.
- a.** Simple random sample (SRS)
 - b.** Convenience sample
 - c.** Systematic sample
 - d.** Cluster sample
- 31** A researcher stands outside a train station and surveys the first 52 people encountered, intending to represent the general population. Identify this sampling method.
- a.** Cluster sample
 - b.** Convenience sample
 - c.** Simple random sample (SRS)
 - d.** Stratified random sample

Multiple choice — medium 15

- 32** A survey about a new dress code is voluntarily filled out online by residents, and only 20% respond. Identify the type of bias most likely present.
- a. Wording bias (leading question)
 - b. Convenience sampling bias
 - c. Voluntary response bias
 - d. Response bias (social desirability)
- 33** A phone survey about university services is conducted using only landline phone numbers, systematically excluding residents without internet access. Identify the type of bias this introduces.
- a. Voluntary response bias
 - b. Response bias (social desirability)
 - c. Undercoverage bias
 - d. Wording bias (leading question)
- 34** Respondents are asked directly, face-to-face, about underage drinking. Identify the type of bias likely to affect their answers.
- a. Wording bias (leading question)
 - b. Voluntary response bias
 - c. Convenience sampling bias
 - d. Response bias (social desirability)

- 35** A survey question reads: "Don't you agree that the wasteful dress code should be changed?" Identify the type of bias in this question.
- a.** Convenience sampling bias
 - b.** Voluntary response bias
 - c.** Wording bias (leading question)
 - d.** Response bias (social desirability)
- 36** A researcher surveys people at a farmers market and generalizes the results to the entire city's population. Identify the type of bias most likely present.
- a.** Response bias (social desirability)
 - b.** Convenience sampling bias
 - c.** Voluntary response bias
 - d.** Undercoverage bias
- 37** A survey about a new policy is voluntarily filled out online by employees, and only 14% respond. Identify the type of bias most likely present.
- a.** Convenience sampling bias
 - b.** Undercoverage bias
 - c.** Voluntary response bias
 - d.** Response bias (social desirability)
- 38** A phone survey about university services is conducted using only landline phone numbers, systematically excluding people who don't check email. Identify the type of bias this introduces.
- a.** Response bias (social desirability)
 - b.** Convenience sampling bias
 - c.** Voluntary response bias
 - d.** Undercoverage bias

- 39** Respondents are asked directly, face-to-face, about tax evasion. Identify the type of bias likely to affect their answers.
- a.** Undercoverage bias
 - b.** Response bias (social desirability)
 - c.** Wording bias (leading question)
 - d.** Convenience sampling bias
- 40** A survey question reads: "Don't you agree that the dangerous parking policy should be changed?" Identify the type of bias in this question.
- a.** Undercoverage bias
 - b.** Voluntary response bias
 - c.** Wording bias (leading question)
 - d.** Response bias (social desirability)
- 41** A researcher surveys people at a shopping mall entrance and generalizes the results to the entire city's population. Identify the type of bias most likely present.
- a.** Undercoverage bias
 - b.** Response bias (social desirability)
 - c.** Wording bias (leading question)
 - d.** Convenience sampling bias
- 42** A survey about a proposed schedule change is voluntarily filled out online by employees, and only 8% respond. Identify the type of bias most likely present.
- a.** Voluntary response bias
 - b.** Response bias (social desirability)
 - c.** Wording bias (leading question)
 - d.** Convenience sampling bias

- 43** A phone survey about school services is conducted using only landline phone numbers, systematically excluding cell-phone-only households. Identify the type of bias this introduces.
- a.** Convenience sampling bias
 - b.** Wording bias (leading question)
 - c.** Undercoverage bias
 - d.** Voluntary response bias
- 44** Respondents are asked directly, face-to-face, about illegal drug use. Identify the type of bias likely to affect their answers.
- a.** Undercoverage bias
 - b.** Convenience sampling bias
 - c.** Response bias (social desirability)
 - d.** Voluntary response bias
- 45** A survey question reads: "Don't you agree that the unfair grading system should be changed?" Identify the type of bias in this question.
- a.** Convenience sampling bias
 - b.** Voluntary response bias
 - c.** Undercoverage bias
 - d.** Wording bias (leading question)
- 46** A researcher surveys people at a train station and generalizes the results to the entire city's population. Identify the type of bias most likely present.
- a.** Voluntary response bias
 - b.** Wording bias (leading question)
 - c.** Convenience sampling bias
 - d.** Response bias (social desirability)

Multiple choice — hard 15

- 47** At a shipping company with 38 delivery routes, a researcher treats each of the 5 major groups as a stratum and takes an independent simple random sample of individuals from every group. Identify this sampling method.
- a. Systematic sample
 - b. Simple random sample (SRS)
 - c. Stratified random sample
 - d. Cluster sample
- 48** At a school with 35 homeroom classes, a researcher randomly selects 4 entire homeroom classes and surveys every individual within them. Identify this sampling method.
- a. Systematic sample
 - b. Convenience sample
 - c. Cluster sample
 - d. Simple random sample (SRS)
- 49** At a school with 57 homeroom classes, a researcher randomly selects 7 entire homeroom classes and surveys every individual within them. Identify this sampling method.
- a. Simple random sample (SRS)
 - b. Cluster sample
 - c. Systematic sample
 - d. Convenience sample

- 50** At a shipping company with 42 delivery routes, a researcher randomly selects 7 entire delivery routes and surveys every individual within them. Identify this sampling method.
- a.** Systematic sample
 - b.** Cluster sample
 - c.** Simple random sample (SRS)
 - d.** Convenience sample
- 51** At a hospital with 60 wards, a researcher randomly selects 3 entire wards and surveys every individual within them. Identify this sampling method.
- a.** Simple random sample (SRS)
 - b.** Convenience sample
 - c.** Cluster sample
 - d.** Stratified random sample
- 52** A school has 46015 enrolled students in total. A researcher surveys a random sample of 162 students to estimate average commute time. Find the size of the population in this scenario.
- a.** 46115
 - b.** 46015
 - c.** 45915
 - d.** 162
- 53** A clinic has 25805 enrolled patients in total. A researcher surveys a random sample of 873 patients to estimate average commute time. Find the size of the population in this scenario.
- a.** 25705
 - b.** 25805
 - c.** 25905
 - d.** 873

- 54** A factory has 20736 enrolled workers in total. A researcher surveys a random sample of 862 workers to estimate average commute time. Find the size of the sample in this scenario.
- a.** 842
 - b.** 882
 - c.** 862
 - d.** 20736
- 55** A company has 44627 enrolled employees in total. A researcher surveys a random sample of 422 employees to estimate average commute time. Find the size of the population in this scenario.
- a.** 44727
 - b.** 44627
 - c.** 422
 - d.** 44527
- 56** A clinic has 44749 enrolled patients in total. A researcher surveys a random sample of 462 patients to estimate average commute time. Find the size of the population in this scenario.
- a.** 462
 - b.** 44649
 - c.** 44749
 - d.** 44849
- 57** A researcher wants to guarantee that every one of 6 very different hospital departments is represented in a patient-satisfaction study. Recommend the most appropriate sampling method.
- a.** Convenience sample
 - b.** Simple random sample (SRS)
 - c.** Stratified random sample
 - d.** Systematic sample

- 58** A pollster has a complete numbered list of all registered voters and wants every voter to have an exactly equal chance of selection. Recommend the most appropriate sampling method.
- a.** Systematic sample
 - b.** Stratified random sample
 - c.** Simple random sample (SRS)
 - d.** Convenience sample
- 59** A mayor wants to fairly represent 4 neighborhoods with very different income levels when surveying about a tax proposal. Recommend the most appropriate sampling method.
- a.** Cluster sample
 - b.** Stratified random sample
 - c.** Convenience sample
 - d.** Simple random sample (SRS)
- 60** A factory manager wants a cheap, logistically convenient way to check product quality by inspecting a few whole production batches out of many. Recommend the most appropriate sampling method.
- a.** Systematic sample
 - b.** Stratified random sample
 - c.** Cluster sample
 - d.** Simple random sample (SRS)
- 61** A quality inspector wants to check items coming off a conveyor belt at a fixed, evenly-spaced interval. Recommend the most appropriate sampling method.
- a.** Cluster sample
 - b.** Systematic sample
 - c.** Simple random sample (SRS)
 - d.** Stratified random sample